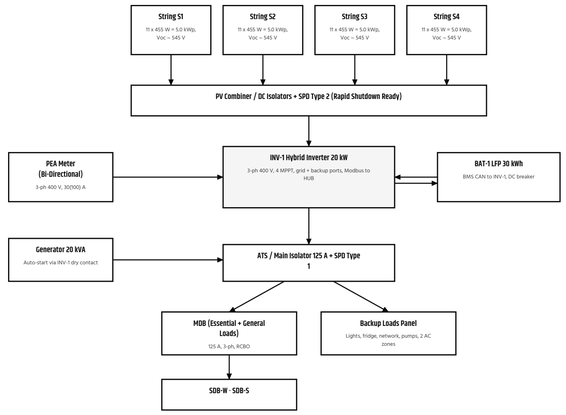
1 **PV ROOF LAYOUT**

SCALE: 3/32" = 1'-0"

SINGLE-LINE DIAGRAM



Earthing: TN-C-S, PV frames bonded, separate DC SPD earth; all to < 5 ohm ring.

ANNUAL ENERGY ESTIMATE

LOAD	KWH/YR	BASIS
Split / VRF air-conditioning	16,400	Connected ~348k BTU/h (~25 kW elec); mountain climate, zoned scheduling
Pool pump (VS) + salt cell	2,600	0.6 kW avg x 10 h + cell
Sauna heater 9 kW	1,700	3 sessions/week incl. pre-heat
Steam generator 9 kW	750	2 sessions/week
Cold plunge chiller	1,100	~3 kWh/day
Heat-pump water heater	900	~2.5 kWh/day
Kitchen, laundry, lighting, plugs, ERV, studio, network, pumps	6,600	~18 kWh/day
Water: well, booster, treatment, irrigation pumps	500	
Total (Without EV)	30,550	~83 kWh/day avg, hot-season days ~110 kWh
Optional EV (15,000 km/yr)	2,400	7.4 kW charger, solar-timed

Peak demand (coincident) about 32 kW: AC 17 kW + sauna 9 kW + range 4 kW + misc; with EV ~40 kW.

OPTIONS (ROUGH THB, 2026)

OPTION	KWP	BATTERY	KWH/YR GEN	SELF-USED	COVERS	COST THB	SAVES THB/YR	PAYBACK
A: 12 kWp + 15 kWh	12	15 kWh	16,560	15,235	50%	570,000	71,000	8.0 yr
B: 20 kWp + 30 kWh (Recommended)	20	30 kWh	27,600	23,460	77%	1,020,000	115,000	8.9 yr
B0: 20 kWp, no battery	20	0 kWh	27,600	15,180	50%	600,000	77,000	7.8 yr
C: Full roof ~125 kWp + 30 kWh	125	30 kWh	172,500	25,875	85%	3,670,000	126,000	29.2 yr

Assumptions: yield 1380 kWh/kWp/yr on ~4 degree roofs; tariff 4.5 THB/kWh (PEA residential incl. Ft); export credit 2.2 THB/kWh only for the first 10 kWp under the PEA/ERC residential rooftop scheme (10-yr contract), surplus above that self-consumed or curtailed (zero-export). Installed cost ~30k THB/kWp incl. hybrid inverter; LFP ~14k THB/kWh.

Recommendation: Option B, 20 kWp on the south-facing planes (master roof + living north half) plus the east wing, with a 30 kWh battery: about 77% of annual use from solar, 9-yr payback, and 6-10 h of backup for essentials during mountain grid outages. The full roof (C) exports mostly unpaid power and never pays back. The north-facing south-wing and studio roofs stay clear (lower yield, sauna/steam vents).

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE

Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
3/32" = 1'-0"

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

SOLAR NOTES

Clamp-on mounts to standing seam: no roof penetrations; keep 0.5 m from eaves for wind uplift.

Permits: PEA grid-connection approval, ERC exemption (< 1 MW), local building notice; engineer-signed roof check.

Tax deduction for residential rooftop solar (2025 measure, up to 200k THB) to be confirmed.

SHEET TITLE

Solar PV Plan + Single-Line Diagram

SHEET

E-102